

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 3 – Low (Pseudo-code Writing)**

Course Code:	CSA0501	Course Title:	Programming in Java
CO Assessed:	CO1 – Precision (BL3,BL4)	Assessment:	Assessment Tool 3 – Pseudo-code Writing
Weightage:		Total Marks:	100
CO1 Statement:	Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)		
Student Name:		Reg. No.:	
Section Batch: /	SLOT B	Date:	22.07.26

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 10 questions, each carrying 10 marks. Total: 100 Marks.
- No negative marking. Un answered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

ANNEXURE

ASSESSMENT TOOL-3

D Student Management System

CLASS Student

PRIVATE : name, rollNO, marks

accept Details()

Calculate Average()

display Result()

END CLASS

Create Student S

S.accept Details()

S.calculate Average()

S.display Result()

STOP

2) Library Book Management

START

CLASS BOOK

PRIVATE : title, author, ISBN, isAvailable

IssueBook()

returnBook()

getDetails()

END CLASS

MAIN

CREATE BOOK b1, b2, b3

b1. IssueBook()

b1. returnBook()

b1. getDetails()

STOP

3) Employee Salary Calculation

START

CLASS Employee

PRIVATE : empId, name, basicPay, designation

CalculateGrossSalary()

CalculateNetSalary()

displayPaySlip()

END CLASS

MAIN

CREATE Employee e

e.calculateGrossSalary()

e.calculateNetSalary()

e.displayPaySlip()

STOP

4) Shape Area Calculator using Polymorphism

START

CLASS Shape

Calculate Area()

CLASS Circle : Shape

Calculate Area()

CLASS Rectangle : Shape

Calculate Area()

CLASS Triangle : Shape

Calculate Area()

MAIN

Create Circle, Rectangle, Triangle

Call Calculate Area()

STOP

5) Vehicle Registration System

START

CLASS Vehicle

display Details()

CLASS Two Wheeler : Vehicle

display Details()

CLASS Four Wheeler : Vehicle

display Details()

MAIN

CREATE Two Wheeler, Four Wheeler

Call display Details()

STOP

6) Bank Account Hierarchy

START

CLASS Bank Account

Withdraw C)

CLASS Savings Account : Bank Account

Withdraw C)

CLASS Current Account : Bank Account

Withdraw C)

MAIN

CREATE Savings Account, Current Account

Call Withdraw C)

STOP

7) Hospital Patient Record System

START

CLASS Patient

generate Bill C)

CLASS IN Patient : Patient

generate Bill C)

MAIN

CREATE IN Patient, Out Patient

Call generate Bill C)

STOP

8) E-Commerce Product Catalog

START

CLASS Product

apply Discount()

CLASS Electronics: Product

apply Discount()

CLASS Clothing: Product

apply Discount()

CLASS Grocery: Product

apply Discount()

MAIN

CREATE Electronics, Clothing, Grocery

Call applyDiscount()

STOP

9) University Staff Hierarchy

START

CLASS Staff

Calculate Allowance()

CLASS TeachingStaff: Staff

Calculate Allowance()

CLASS Lecturer: TeachingStaff

Calculate Allowance()

CLASS ~~Not~~ NonTeachingStaff: Staff

Calculate Allowance()

MAIN

Create Professor, Lecturer, NonTeachingStaff

Call Calculate Allowance()

STOP

10. Smart Home Device Controller

START

CLASS SmartDevice

execute Command (cmd)

CLASS SmartLight : SmartDevice

execute Command (cmd)

CLASS SmartThermostat : SmartDevice

execute Command (cmd)

CLASS SmartLock : SmartDevice

execute Command (cmd)

MAIN

Create SmartLight, SmartThermostat, SmartLock

Call execute Command (cmd)

STOP